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Title

What a Slice-Level Benchmark Certifies without the Pixels: An Audit of Six Public Imaging Datasets

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The journal asks for academic degrees after each name. All four authors have confirmed that none holds a completed academic degree; all four are secondary-school students. The field is therefore left empty rather than filled with a credential nobody holds.

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Author order is the order of the superscripts. There is one first author and one corresponding author; no dual first authorship is requested. ORCID iDs: S.L. 0009-0007-1668-176X; E.T.J. 0009-0009-5602-7599; N.S.V.M. 0009-0004-4744-3877; A.R. 0009-0002-1122-8900.

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The analysis used publicly released label files and the authors' own computers. No institutional imaging archive, clinical system or institutional funding was involved, and the work was not performed under a departmental research program.

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Introduction through Discussion: 2,830 (limit 3,000). Per section, prose excluding headings: Introduction 354 (limit 400), Materials and Methods 800 (limit 800), Results 994 (limit 1,000), Discussion 682 (limit 800). Measured on the compiled PDF as whitespace-separated tokens. The journal's author-instructions page states only the 3,000-word total; its manuscript checklist states the four per-section limits. Both are met. Recount if the text is edited again before upload. Structured abstract: 244 words of prose, 250 counting the four section headings and 251 counting also the word "Abstract" (limit 250). Four thousands-separated figures appear in it; if an extractor splits each on the comma the strictest count is 254. Summary statement: 246 characters (limit 255), one sentence, no abbreviations. Key points: 3 (limit 3). References: 20 cited and printed, of 27 entries retained in the bibliography file (limit 35). Figures: 3 in the main text (limit 6), plus 1 supplemental figure in a separate file. Tables: 4 (limit 4). Abbreviations: 5 declared (limit 10).

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Use of large language models. Large language model assistance was used in two ways: in drafting and editing this manuscript, and in writing the analysis code that produces every number in the Results. The tool was **Claude (Opus 5), Anthropic**, accessed 2026-07-27 to 2026-08-14.

No value reported in the paper was generated by a language model. Every number is a deterministic output of the released code, run on a published label file, and each was re-read from the code's own output before submission. The flagship estimate was reproduced by a second implementation sharing no code with the primary one: run over its own independent holdouts under the same pre-specified protocol, the two families of estimates agree to 0.0003 AUC per slice and 0.005 per patient, and the constant predictor is exactly 0.500 in every draw of both. That reproduction, rather than any assurance about provenance, is what should carry the point, and the code is released so the check can be repeated without the authors.

Other acknowledgments. The authors thank the NYU fastMRI Initiative for releasing the datasets underlying the two audited arms that derive from them, the fastMRI Prostate T2-weighted and diffusion-weighted arms. Two further arms drawn from a fastMRI+ knee release were scored in an earlier round and are withdrawn from this submission. No individual is named, so no written permission to be acknowledged was required.

Data sharing statement

Data generated by the authors or analyzed during the study are available at: `https://github.com/sathvikloke/PhaseDx`, revision `51d3d4bbf38e879af5f13124381f63ba34181e82`, released under a Creative Commons Attribution 4.0 International license.

The first sentence above is one of the five statements the journal requires verbatim, with the citation to the data supplied as the policy directs; the revision identifier is given because the Algorithm and Code Transparency policy requires one. That revision is the commit whose code produced every value reported in this manuscript; any later commit on the same branch edits manuscript prose only and changes no reported number. The authors undertake to maintain the repository and link for the five years the policy specifies, and will mint an archival persistent identifier for the revision of record on acceptance. What follows describes the restrictions on reuse, as the policy also asks.

The archive contains the audit code, the per-benchmark output payloads, the label-table preparation scripts, the frozen-holdout re-analysis, a second implementation of the intracranial hemorrhage unit-collapse computation sharing no code with the first, and the assembler that builds the cross-study comparison. It does not contain, and cannot contain, any benchmark's pixel data. It also holds material from other parts of the larger study that this submission does not report; see the cover letter.

The label files analyzed here were generated by third parties, not by the authors, and each carries its own terms. Five of the six benchmarks are public releases obtained under their own published terms. The worked-example label files are

NYU fastMRI releases, available by application to the NYU fastMRI Initiative (`https://fastmri.med.nyu.edu`) under a data sharing agreement that permits use in academic publications but forbids onward distribution of the dataset or of files derived from it; those files are therefore not redistributed in the archive, and the code that regenerates every reported value from a reader's own authorized copy is released instead. No identifiable or private participant-level information was held by the authors. The flagship label file carries no age or sex, which is why neither is reported for it; two audited label files do carry demographic or clinical columns, and Table 4 names them.

Conflicts of interest

All four authors declare no conflicts of interest. No author has a financial interest in the subject matter of this manuscript, and no author has any affiliation with, or financial involvement in, any organization with a financial interest in it. No author holds any position, fiduciary or otherwise, with any organization that produced, funds, distributes or maintains any of the benchmarks audited here, or with any organization whose published results are used as comparators. Each author will submit a completed ICMJE Disclosure of Potential Conflicts of Interest Form.

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